

1 Fields of Application BDE

1.1 Overview

The goal of the following description is to present the many collection and processing options offered by HYDRA shop floor data collection (HYDRA-BDE) and machine data collection (HYDRA-MDE).

1.2 Basics of shop floor data collection

The postings of the HYDRA-side production data collection are based on the components machine, order and person. Depending on the approach and the main focus of the individual use of HYDRA, different components might play a prominent role or they might be unnecessary and therefore as far as possible disabled.

Machine/workplace – where does production happen?

Logging an order or employee on or off is basically machine or workplace related. Logging on to a machine answers the question of where a BDE posting takes place. BDE postings can basically be executed only on those machines or workplaces which have been configured accordingly in the HYDRA master data.

The names *machine* and *workplaces* are synonymous in HYDRA. In HYDRA-MDE, the expression "machine" is favored. In HYDRA-BDE, the name "workplaces" is favored.

Order/operation – what is produced?

All tasks performed by staff at machines are based on orders. Logging the order or its operation on answers the question what is done or which activity is executed.

HYDRA differentiates between different order types to classify orders according to their utilization. Different control information that is decisive for managing the orders is defined for each order type.

HYDRA provides the following order types by default. Further order types can be created according to the customer's requirements:

- Production order

A production order generally relates to the article/item and is characterized by a target quantity as well as a completion date.

- Overhead cost order
An overhead cost order (e.g. cleaning of the workplace) normally only refers to cost-accounting.
- Capacity order
A capacity order has only been designed for planning purposes (assignment of capacities) within HYDRA shop floor scheduling. You usually change capacity orders into production orders at a later stage.
- Project order
A project normally is unique. A project order combines the project steps that are carried out during a project.
- Maintenance order
A maintenance order has been designed for the planning or recording of maintenance measures.

In general, orders are created in a higher level system (e.g. ERP system) and transferred to HYDRA using an interface. Nevertheless, orders can also be created manually in HYDRA. Planning of orders (in the ERP system or HYDRA) depends on precise requirements.

Orders are mostly multi-level and divided into several operations to be processed on different workplaces and machines. The order and operation are generally logged on to the shop floor terminal at the individual workplace. Data can only be posted onto operations at shop floor terminals if a corresponding pool of orders exists in the HYDRA database.

The terms 'order' and 'operation' are mostly synonymous in the handbook descriptions. Usually, when posting functions on the terminal are being described, the operation is meant, and documentation on HYDRA shop floor data collection also emphasizes operation data. Only in certain HYDRA descriptions is the 'Order' used as an umbrella term for the whole multi-level production order.

Person - who is working?

The relation to the machine and operation is established in HYDRA by logging staff on and off at shop floor terminals. The work confirmations of employees form a basis for the calculation of personal expenses and for performance determination. With the logon or logoff of staff, you are informed who is working, where the person is working and what is done.

The relation of a posting to an employee is mostly optional. The logging on and off of production personnel is subject to fundamental decisions which are made before an MES system is introduced. HYDRA usually only allows those employees to log on, who are stored in the HR master data of the HYDRA database and authorized in the company.

1.3 Presentation of recording results

The components of data collection - machine, order, person, - are summarized to the greatest possible extent in logical dialogs. Consequently, an employee may, for example, log on and off to and from a terminal together with an order.

Decisions as to the design of individual dialogs on the shop floor terminals depend on the environment in which it is used, the industry sector, the organization, the range of items, the order structure and the machinery involved.

The data collection options in HYDRA represent a range of exemplary uses, which are supported by the modules of the standard system. A single installation requires only one selection of the possible configuration options. Individual HYDRA settings are finalized by the HYDRA user, usually together with an MPDV consultant, after successful installation and in the course of customizing.

1.3.1 Time recording

"Posting events" on shop floor terminals form the basis for the posting of particular machine times to an operation. A time posting is initiated by the posting event "logon" and completed logically by the posting event "logoff" or "interruption". This basic principle applies to all order and personnel posting events.

The time posting of shop floor data collection primarily occurs in two different time accounts, one for machine scheduling duration and the other for labor utilization.

The **machine time** is specified by the time interval between the logon and logoff of an operation. The machine scheduling time is harmonized with the shift calendar of the machine. Planned shift breaks are not included in the time interval calculation.

Personnel deployment represents the total of all labor times for each operation. This period is determined by the time interval between logon and logoff of the user or users. The basis for personnel postings is, once again, the shift model of the workplaces and the breaks it contains. If employees are processing more than one order simultaneously, then the HYDRA system carries out a proportional calculation of the operating time for each operation.

For further details on this please refer to the section entitled "posting of times".

The results are documented in "log records" that are generated automatically due to posting events.

1.3.2 Recording quantities

Quantities are separately maintained in HYDRA for each data collection level: machine, order, and person:

- **Machine related**
e.g. for display of the shift performance in the machine efficiency report
- **Operation related**
e.g. for display of the target-actual comparison in the order information
- **Personnel related**
e.g. as the basis for incentive pay based on quantities

Depending on the data collection type, quantities are either entered **automatically** (MDE/BDE terminal with counter connection) or must be **entered manually**.

Manual quantity postings are carried out in the entry dialog at the end of personnel or order processing or as partial confirmation/upload during order processing. Depending on the objective, the entry dialogs *Interrupt operation*, *Logoff operation*, *Logoff person* and *Partial confirmation/upload* are configured for the input of quantities. Quantities that are recorded manually are always posted onto the operation that is logged on. Posting onto the person realizing the posting depends on the posting type or the respective parameter settings configured in HYDRA.

Along with manual entry of quantities, HYDRA also supports **automatic entry of quantities** with active machine data collection. With this type of entry of quantities, a continuous quantity posting to the machine, and to the logged on operation and the logged on employees takes place at cyclical intervals.

All OPs and employees logged on to the machine are automatically assigned quantities. The quantities are computed according to the partitioning/cavity that is respectively defined for the operation and is posted onto the operations or persons.

HYDRA provides different accounts for quantity posting. The accounts that are predominantly used are yield and scrap. The two fields are provided appropriately during data collection. Scrap quantities can be classified by a scrap reason. The available scrap reasons are configured in HYDRA.

Other quantity accounts are *Rework* and *Open quantity*. These are usually only used specifically in form of customer-specific scenarios; the fields are provided as part of the customizing process.

1.3.3 Options of Machine Data Collection

If you connect a machine to a shop floor terminal, the automatic data collection is possible.

- Collection of production times and downtimes
- Direct assignment of malfunction reasons by collecting operating signals from machine control systems
- Piece number recording for yield, total quantity and scrap

Every workplaces and machine is assigned a shift model in the HYDRA configuration. The shift model determines the capacity of the machine and gives the shift calendar a reference time for the performance of a machine. All machines for which recording of standstills is carried out in HYDRA, have a definite machine status at all times during the shift. The total of all production and downtimes during a shift gives exactly the shift standard time.

Definition of downtime reasons

The determination of the relevant machine statuses forms the basis of machine monitoring by means of machine signals. Together with the "production" status all data collection relevant downtime reasons for the machine are created in HYDRA. The user may configure them individually.

Classification in HYDRA resource performance accounts (RPA)

HYDRA resource performance accounts are a system of time accounts, consisting of 12 accounts. They group together similar downtime reasons in a single account (e.g. all technical disturbances go in the "disturbance-caused interruptions" (DCI) resource performance account). During data collection, accumulating times are posted to the resource performance account to which the current machine status is assigned in the system configuration.

The standard definition of the HYDRA resource performance accounts is as follows:

No.	Acronyms	Description
1	SUT	Secondary utilization time

No.	Acronyms	Description
2	DCI	Disturbance-caused interruption (= technical interruption)
3	LCI	Logistics-caused interruption (= organizational interruption)
4	SCI	Staff caused (personnel related) <i>interruption</i>
5	IMN	Idle mode, not scheduled (e.g. repairs)
6	IMS	Idle mode, scheduled (e.g. maintenance)
7	SET	Setup
8	STA	Startup
9	U8	Free for user (e.g. trial production)
10	U9	Free for user
11	MUT	Main utilization time; "Production"
12	BKS	Neutral times, e.g. free, breaks etc.

If recording of downtimes is not required or if the recording of downtime reasons gives no clear result (e.g. for HYDRA group workplaces), then time recording generally takes place with *Production* status and is represented in the MUT main utilization time performance account.

The HYDRA resource performance accounts are also kept relating to operations and persons. Machine scheduling times (machine duration) for operations and personnel result from the following calculation:

Scheduling time = total (RPA 1 .. RPA 11).

Automatic recording of number of pieces using machine signal connection

Recording a cyclical signal from the machine enables the recognition of downtimes, and by counting the recorded cycles it is also possible to record the number of pieces produced.

To ensure an accurate determination of the number of pieces, HYDRA supports the multiplication of recorded cycles with multiple accesses per cycle. "Partitioning" (also called cavity) is a tool specific parameter, which is transferred to the terminal when an operation is logged on.

Automatically recorded data are posted as machine performance in HYDRA and simultaneously assigned to the operation and personnel currently logged on.

Collection	operation-related	Personal
actual quantities	X	X
machine times	X	X
Labor utilization	X	X